Program : **Diploma in Chemical Engineering**	
Course Code : **2071**	Course Title: **Chemical Process Principles**
Semester : **2**	Credits: **3**
Course Category: **Engineering Science**	
Periods per week: **3 (L:3 T:0 P:0)**	Periods per semester: **45**

Course Objectives:

- To orient the students towards chemical engineering as a profession.
- To enable the students to apply gas laws and material balance to solve gas handling and material balance problems in process industries.

Course Pre-requisites:

Topic	Course code	Course name	Semester
Knowledge of basic Mathematics		Mathematics	1
Knowledge of basic Physics		Applied Physics I	1
Knowledge of basic chemistry		Applied Chemistry	1

Course Outcomes:

On completion of the course student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Interpret the fundamental concepts of chemical engineering.	10	Applying
CO2	Apply gas laws to solve gas or vapour handling problems.	14	Applying
CO3	Apply material balance without chemical reactions in chemical process calculations.	12	Applying
CO4	Apply material balance with chemical reactions in chemical process calculations.	9	Applying

CO – PO Mapping

Course Outcomes	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7
CO1	3						
CO2	3						
CO3	3						
CO4	3						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcome	Description	Duration (Hours)	Cognitive Level
CO1	**Interpret the fundamental concepts of chemical engineering.**		
M1.01	Identify the roles of a chemical engineering technician.	1	Understanding
M1.02	Outline process plant operations.	1	Understanding
M1.03	Summarise the fundamental concepts of Stoichiometry.	4	Understanding
M1.04	Apply the fundamental concepts of Stoichiometry to solve chemical engineering problems.	4	Applying
Contents: History of chemical engineering as a discipline - Greatest achievements in Chemical Engineering - Relate Chemistry and Chemical Engineering - Role of a chemical engineering technician – Compare the duties of a chemist and chemical engineering technician - List the major chemical industries in India and Kerala with their raw materials and products. Overview of batch process –Transition from batch to continuous process – Unit operations – Unit process. Gram Atom - Gram mole - Atomic Weight - Molecular Weight – Methods of Expressing concentration of solutions – Molarity - Normality - Molality - Methods of expressing composition of mixtures - Weight Fraction - Weight Percent - Mole Fraction - Mole Percent - Volume Fraction -Volume Percent - Conversion of Weight% to Mole%- Conversion of Mole% to Weight%. Simple problems based on above relations.			
CO2	**Apply gas laws to solve gas or vapour handling problems.**		
M2.01	Summaries the basic gas laws applicable to chemical engineering.	2	Understanding

M2.02	Apply the basic gas laws to solve gas / vapour handling problems.	4	Applying
M2.03	Interpret the relationship between partial pressure fraction and mole fraction.	2	Understanding
M2.04	Estimate the average molecular weight and density of gas mixtures.	2	Understanding
M2.05	Apply the relationship between partial pressure, mole fraction and average molecular weight to solve gas handling problems.	4	Applying
	Series Test – I		

Contents:

Basic gas laws - Boyles law - Charles law – Ideal gas equation – Estimate the value of R in different units - Dalton's law - Amagat's law.

Solve simple problems based on above laws.

Relationship between partial pressure fraction and mole fraction of a component gas to total pressure.

Average molecular weight of gas mixture – Density of gas mixture.

Solve simple problems based on partial pressure fraction, mole fraction, average molecular weight and density of gas mixture.

CO3	**Apply material balance without chemical reactions in chemical process calculations.**		
M3.01	Summarise the law of conservation of mass.	2	Understanding
M3.02	Summarize the classification of material balance problems and outline a procedure for solving material balance problems.	2	Understanding
M3.03	Outline the unit operations in chemical process industries.	2	Understanding
M3.04	Solve material balance problems for the unit operations in chemical process industries.	6	Applying

Contents:

State the Law of conservation of mass

Classification of material balance problems – Procedure for solving material balance problems.

Outline the unit operations - Distillation – Evaporation – Absorption – Extraction – Drying – Filtration – Mixing – Crystallization.

Solve simple material balance problems for above unit operations.

CO4	**Apply material balance with chemical reactions in chemical process calculations.**		
M4.01	Summarise Stoichiometric equation, Stoichiometric coefficient, Stoichiometric ratio and Stoichiometric proportion.	1	Understanding
M4.02	Summarise Limiting reactant, Excess reactant and Percentage excess.	1	Understanding
M4.03	Compute percentage conversion, yield and selectivity	1	Understanding
M4.04	Solve problems on material balance with chemical reactions.	5	Applying
M4.05	Illustrate bypass, recycle and purge operations in process industries.	1	Understanding
	Series Test – II		

Contents:

State: Stoichiometric equation - Stoichiometric coefficient - Stoichiometric ratio - Stoichiometric proportion.

Limiting reactant - Excess reactant - Percentage excess - percentage conversion - yield - selectivity.

Solve the problems based on - Stoichiometric proportions of reactants and products - Limiting reactant - Excess reactant - Percent excess - Percentage conversion - Yield - Selectivity - Combustion of fuels - Percentage of excess air.

Illustrate the process operations line Bypass, recycle and purge.

Text / Reference

T/R	Book Title/Author
T1	Stoichiometry and process calculations, K V Narayanan & B Lakshmikutty
R1	Stoichiometry, B. I. Bhatt & S. M Vora
R2	Introduction to process calculations, K A Gavhane
R3	Chemical process Principles, Hougen, O. A Waston K. M, Regatz. R. A

Online Resources

Sl.No	Website Link
1	https://nptel.ac.in/courses/103/103/103103165/